Problem Set 2: Macroeconomics and Inequality

Juan Zurita and Zhuotong Liu
Programming and Numerical Methods for Economics—ECNM10115
The University of Edinburgh

Due to Thursday, 09-Oct, 14:00 pm

2025

For this problem set we will use the 2011-2012 wave of the Ugandan National Panel Survey. The survey is under the umbrella of the ISA-LSMS surveys and offers nationally representative household data on consumption, income, wealth and other key variables in Uganda. The file *UNPS_1112_PS2.xls* contains the data while the file *variables_description_UNPS_1112_PS2.xls* contains a description of the variables. Import the data and answer the following exercises.

Exercise 1. Exploring the data (25 points).

a. Are there duplicate households in the data? Check if there are repeated observations in the unique household identifier variable. How many observations are in this dataset?

b. Present a summary statistics for the following variables: *head_gender, head _age, familysize, consumption, income, wealth*. Summarize your results in two lines. Specifically, note if there are any missing observations or potential outliers for any of the variables.

c. Using the *head_gender* variable, create a dummy variable for household head being female (1=female, 0=male). What is the percentage of households with a female head?

d. Using the groupby method, calculate the average consumption, average household size, and average age of the household head for households with a male head versus those with a female head. Do you observe any significant differences between the two groups?

Exercise 2. Inequality in Uganda (50 points).

a. Create the variables log_c, log_inc, and log_w as the log of consumption, income, and wealth, respectively. Plot the distributions of log consumption and log income on the same graph. Do the distributions resemble any known distributions? Is inequality higher in consumption or income?

b. A commonly used statistic to measure inequality is the variance of the logs. Compute the variance of the log of consumption, of the log of income, and of the log of wealth. How do these measures of inequality in Uganda compare to the same measures of inequality in the United States? Use table 3, column 5–PSID in De Magalhães, L., & Santaeulàlia-Llopis, R. (2018) for the comparison.

c. Measuring between rural and urban inequality in Uganda. Compute the average consumption, income, and wealth for rural and urban areas separately (groupby). Are the differences between the two areas large?

d. Measuring within rural and urban inequality in Uganda. Compute the variance of the log of consumption, income, and wealth for rural and urban areas separately.

e. Compute the Gini coefficient for consumption, income, and wealth. Compare these values with the Gini coefficients in the United States—table 3, column 5–PSID in De Magalhães, L., & Santaeulàlia-Llopis, R. (2018)

f. Compute the share of the wealth held by the bottom 50 percent. Compute the share of the wealth held by the top 10%, 5%, and 1%.

g. Although there has been significant debate on inequality in recent years, the focus has largely been on rich countries. Based on the results of this exercise, discuss whether inequality in Uganda is relatively larger compared to that in rich countries.

h. Previous studies on income inequality in Africa have often relied on consumption measures to estimate income inequality. For example, see Alvaredo & Gasparini (2005). Discuss the advantages and disadvantages of using consumption measures to study income inequality.

Exercise 3. The lifecycle of male vs female head households in Uganda (25 points).

Before plotting, you may want to exclude age groups with few households, such as those above 80 years old or below 18 years old. Additionally, consider grouping the ages into bins to create smoother plots.

Then, using seaborn lineplot with the argument *hue='female'*, or any other variable to distinguish the gender of the head:

a. Plot the lifecycle of the log of consumption for households with a male head and for households with a female head.

b. Redo the same plot but for the log of income (i) and for the log of wealth (ii).

c. What are the differences in the lifecycle of consumption, income, and wealth between households with male and female heads? Please comment on your findings.